

Engineering of Data Analysis

Lecture 1

Outline

Big data, data mining, machine learning, etc.

Data analytics

- Descriptive vs. predictive
- Data analysis models

The Big Data Era



Big Data is changing every aspects of our lives.

From sports to entertainments, from healthcare to autonomous cars.

Image source: Dursun Delen & Sudha Ram (2018) Research challenges and opportunities in business analytics, Journal of Business Analytics.

Big Data?

The size of digital data is growing exponentially.

Forecast [IDC'25]:

~174 ZB of data created in 2025; on track for 230–240 ZB in 2026 and ~528 ZB by 2029

Data contains value and knowledge.

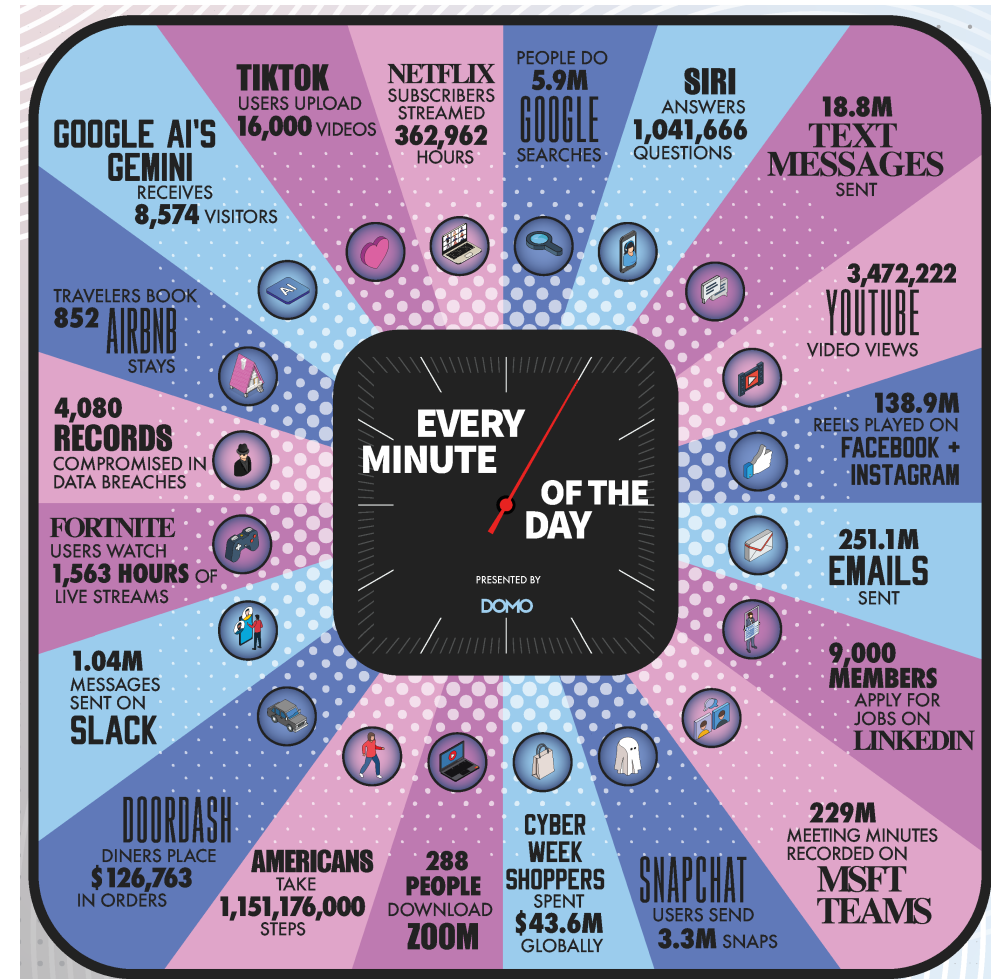


Image source: <https://www.domo.com/learn/infographic/data-never-sleeps-12>

Big Data and Data analysis

But to extract the knowledge (and value), data needs to be:

- Collected
- Stored (and Managed)
- Analyzed

Data analysis

Analytics The science that analyse raw data to extract useful knowledge (patterns) from them.

This process can also include data collection, organization, pre-processing, transformation, modelling and interpretation.

Data analysis: an old idea

The term “business intelligence” was used by the first time by Richard Millar Devens in 1865 to describe how the banker Henry Furnese achieved an advantage over competitors by:

- Building a network to collect information across Europe (markets, politics, wars).
- Receiving critical news before competitors; and
- Acting on that information quickly to make profitable decisions.

This is probably the first study of using **data analysis for commercial purposes**.

Data analysis: an old idea (2)

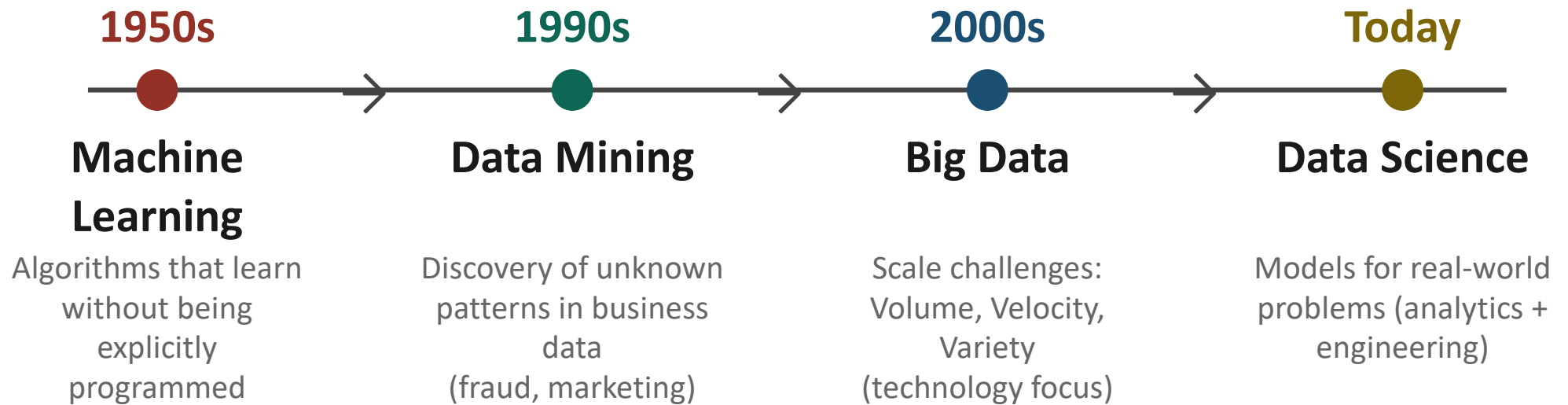
In 1880, the US Census Bureau estimated that it would take it more than 10 years to crunch all the data collected in the 1890 census, meaning that results would be outdated by the 1900 census data.

To solve the problem, Herman Hollerith produced the Hollerith Tabulating Machine, that reduced the time to three months. First **automated** data processing system.

The company he founded became IBM.



From data mining to data science



Key distinction: Data science focuses on creating **models** for real-world problems. Big data focuses on the **technology and infrastructure** to process data at scale. This course is about the engineering side.

Adapted from: Moreira, Carvalho & Horvath (2018)

The 3 Vs of Big Data

Volume

The sheer amount of data.
"By 2026 the global
datasphere
will grow to 221 zettabytes."

1 billion Facebook users in a single day

Velocity

The speed at which data
arrives and must be
processed
in real time or near-real time.

Financial tickers, fraud detection

Variety

Different sources and formats:
tabular, text, images, video,
sensor data, logs.

Combining CRM, social media, IoT

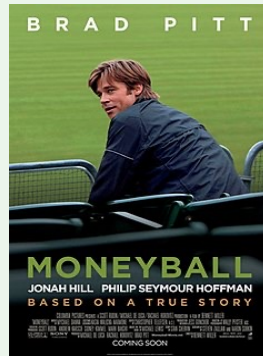
Engineering implication: Each V poses a different systems challenge — storage architecture, stream processing, and data integration.

Data analysis in the real world

Moneyball (2011)

Using data to challenge conventional expertise

Oakland A's general manager used statistical analysis to build a competitive baseball team on a limited budget — proving that data-driven decisions could outperform traditional scouting and intuition.



The Big Short (2015)

Finding hidden patterns others missed

A handful of investors analyzed mortgage-backed securities data and discovered systemic risk that the entire financial industry had overlooked — demonstrating the power of independent data analysis to uncover critical insights.



The lesson: Data analysis creates value when it reveals patterns invisible to conventional approaches. But getting from **raw data** to **actionable insight** requires engineering — the right storage, the right frameworks, the right processing models. That's what this course is about.

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- **Descriptive vs. predictive**
- Data analysis models

Data Analytics

Given lots of data (including tabular, images, video, etc.)

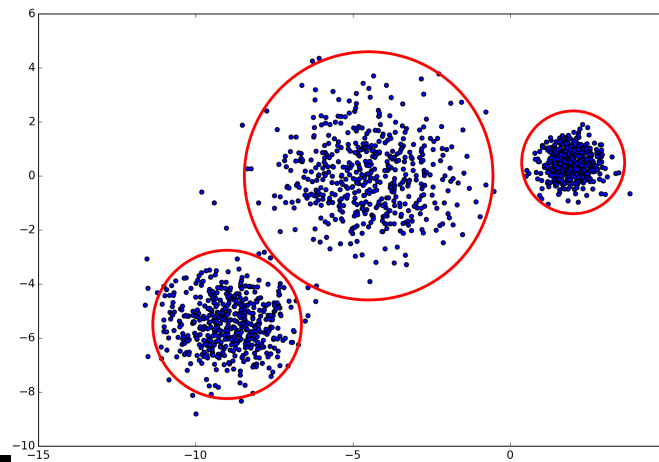
Discover patterns and models (knowledge) that are:

- **Valid**: hold on new data with some certainty
- **Useful**: should be possible to act on the item
- **Unexpected**: non-obvious to the system
- **Understandable**: humans should be able to interpret the pattern

Data Analytics Taxonomy

Descriptive analytics

- Find human-interpretable patterns that describe the data.
- Unsupervised approach – the interpretation is unknown.
- The result of a given method or technique is obtained directly by applying an algorithm to the data.
 - Example: Simple aggregations such as average or complex algorithms such as clustering



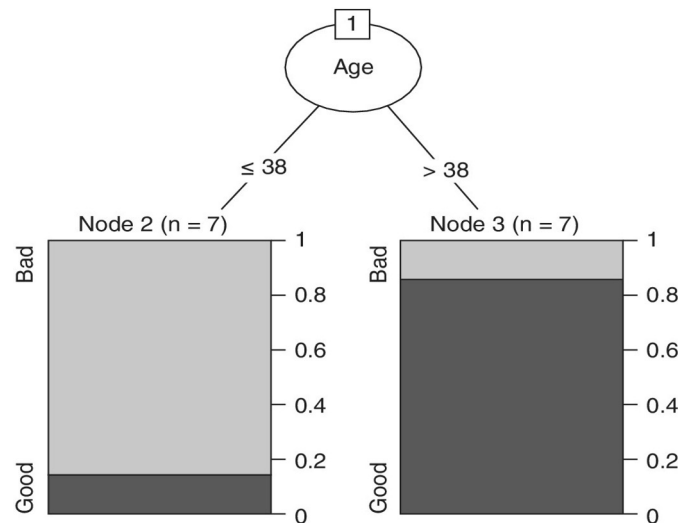
Data Analytics Taxonomy (2)

Predictive methods

- Extract models to predict unknown or future values.
- Supervised approach – the data is labelled.
- The result of applying an algorithm of a predictive method to given data, is typically a model.
 - Example: Recommender systems
- The model is used for predicting the value of new data points.

But what is a model?

A **model** is a generalization obtained from data that can be used afterwards to generate predictions for new given instances.



Name	Age	Educational level	Company
Andrew	55	1	Good
Bernhard	43	2	Good
Carolina	37	5	Bad
Dennis	82	3	Good
Eve	23	3.2	Bad
Fred	46	5	Good
Gwyneth	38	4.2	Bad
Hayden	50	4	Bad
Irene	29	4.5	Bad
James	42	4.1	Good
Kevin	35	4.5	Bad
Lea	38	2.5	Good
Marcus	31	4.8	Bad
Nigel	71	2.3	Good

Predictive vs. Descriptive methods

Descriptive methods can also be used for predictive processes.

Examples:

- Clusters created by clustering can be used to assign data points to a group.
- Moving averages can be used to (try to) predict future values.

Implications for data analysis engineering

Descriptive methods are **one-step processes**:

- From the data, an algorithm is used to derive the (human-interpretable) result.

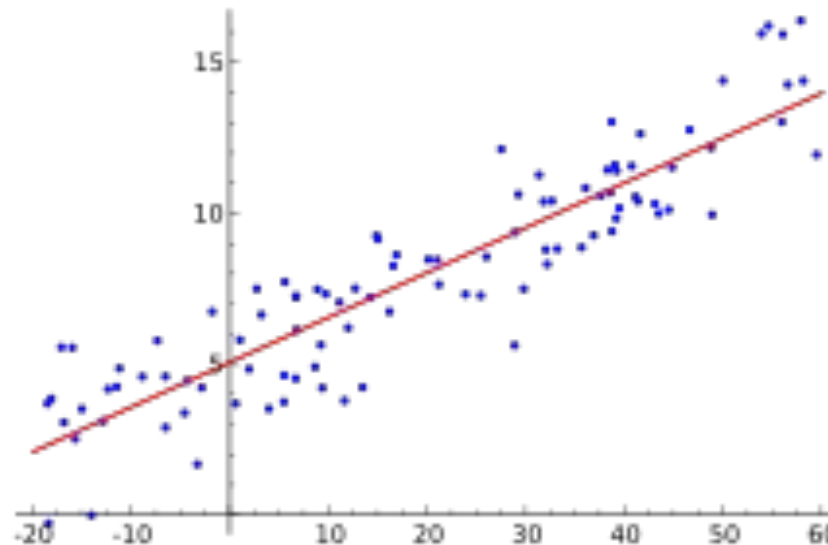
Predictive methods are **two-step processes**:

- From the data, an algorithm is used to derive the model – training step.
- With the model and a new data sample, an algorithm is used to classify the data sample – inference step.
- The training step is typically much more computationally demanding than the inference step.

Parameters and hyper-parameters

Model **parameters** are values that are set by a modelling or learning algorithm in its internal procedure.

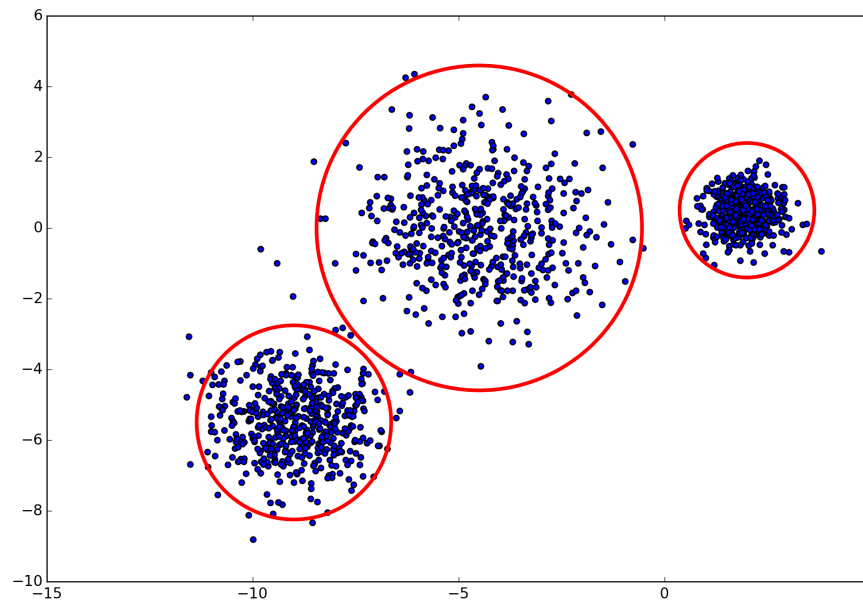
- E.g. the slope parameter of multivariate linear regression



Parameters and hyper-parameters (2)

Hyper-parameters are values set by the user, or some external optimization method.

- E.g. the number of clusters in k-means



Implications for data analysis engineering

The value of hyper-parameters is often unknown. Example:

- The number of clusters when doing clustering is unknown.
 - Need to explore different values and determine the “best” value.
 - This can be done **independently** => in **parallel**.

The best value for model parameters is often unknown. Example:

- The parameters of a neural network.
 - Can create different models to determine the “best” one.
 - This can be done **independently** => in **parallel**.

=> Both are embarrassingly parallel — prime candidates for distributed computing.

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Data processing: batch vs. stream processing

Data analysis/processing tasks consist in executing computations over data.

Two main data processing models:

Batch processing consists in performing computations over a dataset that is available when the computation starts.

Stream processing consists in doing continuous computations over data that is being created/received.

Batch processing: Exploratory & Interactive Analysis



The analyst explores the full dataset before any computation; the entire dataset must exist before the notebook runs.

Jupyter Notebook – customer churn EDA

- A data scientist loads 2M rows of e-commerce transactions into a pandas DataFrame, plots purchase-frequency distributions, computes RFM scores (Recency, Frequency, Monetary value — how recently, how often, and how much each customer bought), and iterates on visualisations cell by cell to decide on a churn model

See also: Ad-hoc SQL on Snowflake to investigate a conversion rate drop; seasonal pattern analysis on 30 years of climate data in R

Batch processing: Business Intelligence & Reporting



BI tools query pre-computed snapshots rather than live data; results are fast because the heavy aggregation was done in an earlier batch job.

Tableau / Power BI sales dashboard

- Every morning, a scheduled job runs aggregation queries over last night's data warehouse snapshot; Tableau refreshes a regional sales dashboard that 200 managers consult at 9 am — results are pre-computed, not live

See also: Automated weekly PDF report of supply-chain KPIs emailed to buyers; nightly OLAP cube rebuild for slice-and-dice margin analysis

Batch processing: Repetitive Tasks in Larger Systems



Recurring batch jobs are embedded as steps inside larger data pipelines; they run unattended on a schedule and feed downstream systems.

Weekly ML model retraining – Spotify Discover Weekly

- Every Monday, Spotify's pipeline retrains its collaborative filtering model on the past week's listening events (~30 billion streams); the new model artefact is evaluated, versioned, and deployed — Discover Weekly playlists are generated fresh each Monday morning for 600M users

See also: *Nightly ETL ingesting web server logs into Redshift via Airflow; post-ingestion data quality checks with Great Expectations*

Batch processing: GPU-intensive Workloads



GPU workloads process large volumes of data in parallel; they run as batch jobs because the full dataset (images, tokens, frames) must be available before processing begins.

Medical imaging batch inference – radiology triage (Aidoc)

- Aidoc runs a PyTorch model overnight on the day's accumulated CT and chest X-ray scans (hundreds of thousands of images per hospital network); each scan is scored for critical findings on a GPU cluster, and radiologists receive a prioritised worklist by morning — inference cannot start until the scans are complete

See also: *CGI rendering of a VFX sequence distributed as independent frames across hundreds of GPU nodes*

Batch processing: Distributed Data Processing



Data too large for one machine is split across a cluster; each node processes its shard in parallel, and results are combined — no single machine could hold or process the full dataset.

Customer lifetime value segmentation – retail Spark job

- A retailer runs a weekly Apache Spark job over 5 years of transaction logs (10 TB on S3); the job aggregates per-customer RFM features across a 50-node cluster, applies a k-means model, and writes CLV segments back to the data warehouse for targeted marketing — the full history is needed to compute lifetime value

See also: *UK Biobank whole-genome sequencing pipeline for 500k patients on a cloud Spark cluster*

Stream processing: Edge & Embedded Infrastructure



Processing happens on the device itself; cloud connectivity is either unavailable, too slow, or introduces unacceptable latency for safety-critical decisions.

Autonomous vehicle perception – Tesla FSD chip

- Tesla's Full Self-Driving chip processes 8 camera streams at 2,300 frames/sec entirely on-board; lane detection, object classification, and path planning all run on the vehicle's neural-network accelerator — a cloud round-trip of even 50 ms would be incompatible with safe braking decisions at highway speed

See also: *Wind turbine bearing monitoring triggering shutdown in milliseconds via on-device DSP*

Stream processing: Broker-based & Cloud Infrastructure



A message broker decouples producers from stream processors; stateful computations (windowed counts, ML scoring) run in a cloud or datacenter cluster with low but non-zero latency.

Payment fraud detection – Kafka + Apache Flink (Visa)

- Visa processes 65,000 transactions/sec; each event flows through Kafka into a Flink stateful job that maintains per-customer rolling spend windows and applies an ML scoring model — a fraud decision is returned in under 100 ms, before card authorisation completes; batch processing yesterday's data would be far too late

See also: *Google Maps updating traffic overlays every 2 min via Pub/Sub + Dataflow; Twitter/X computing trending hashtags over a 30-min sliding window via Kafka + Storm*

Programming data analytics

Different from general-purpose programming

- Performed by non-CS specialists;
- Apply sequences of methods/techniques for processing data;
- Require exploring parallelism easily – GPUs, distributed processing.

Multiple frameworks for data processing/data analytics exist.

Programming data analytics

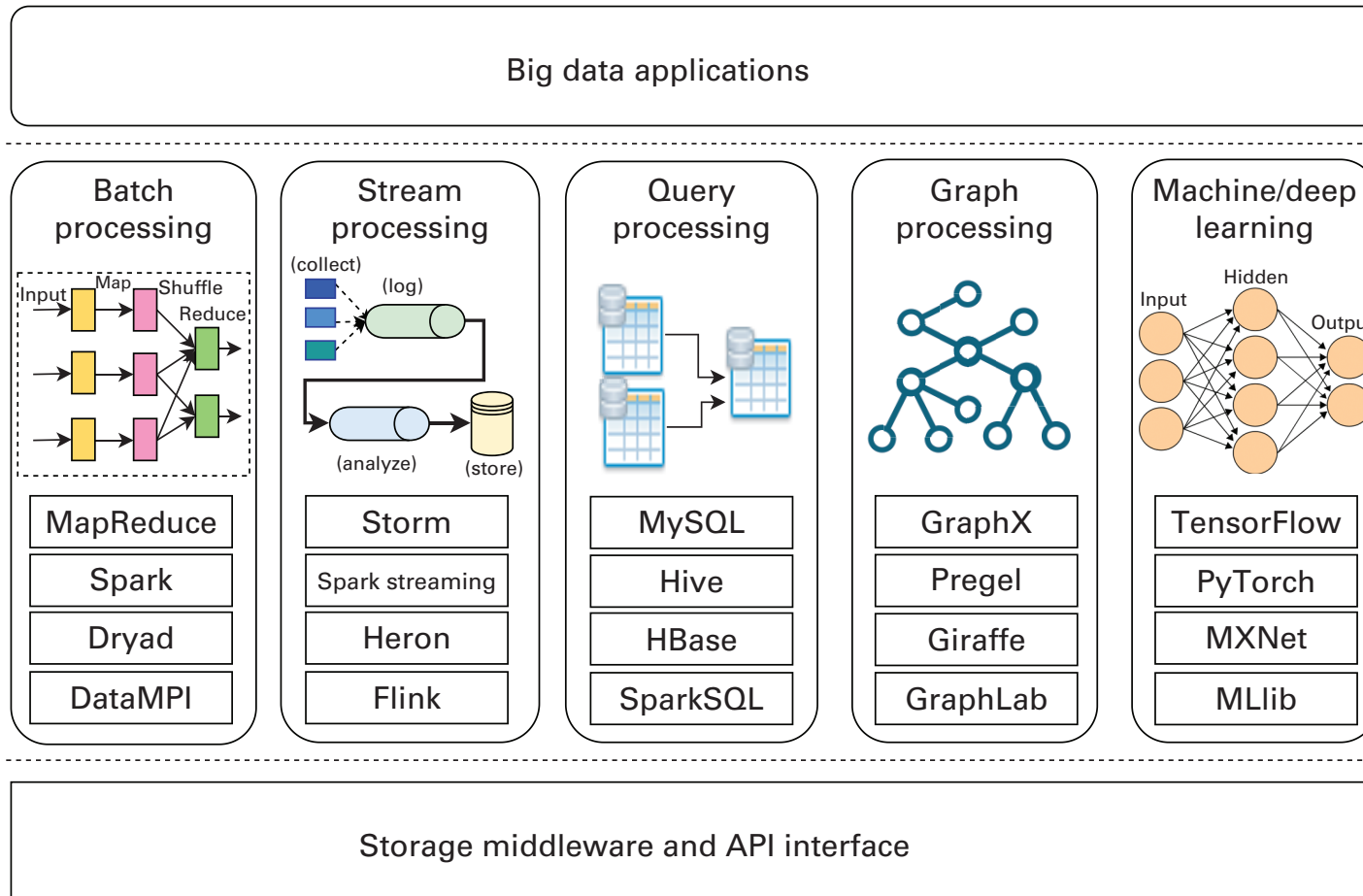


Image source: High-Performance Big Data Computing, MIT Press.

References

Dhabaleswar K. Panda, Xiaoyi Lu and Dipti Shankar. High-Performance Big Data Computing. The MIT Press

João Moreira, Andre Carvalho, and Tomás Horvath. A General Introduction to Data Analytics. John Wiley & Sons, 2018. Chapter 1.

Images and examples from these books.